



Progress Report & First Results

CAR SEGMENTATION

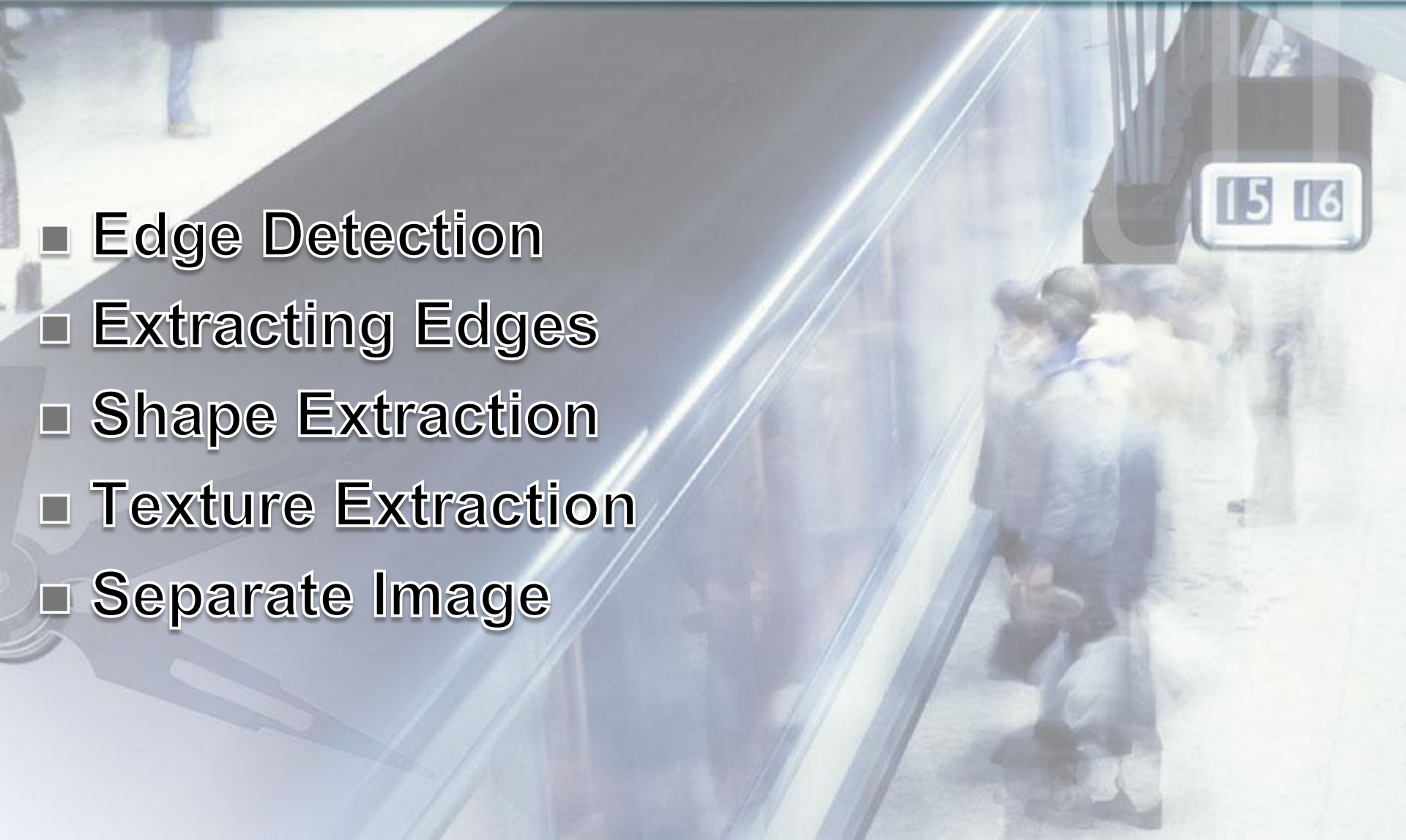
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time management

First Steps

- Edge Detection
- Extracting Edges
- Shape Extraction
- Texture Extraction
- Separate Image



Edge Detection

- Edge detection is a fundamental concept for segmentation. This is easy not only to understand its principle but also to implement.



Method:



```
im = imread('Auto2.jpg');  
%figure, imshow(im);  
  
figure,  
im2= im2double(im);  
subplot(2, 2, 1),imshow(im2); title('Iam2Double');  
  
im3= rgb2gray(im2);  
subplot(2, 2, 2),imshow(im3); title('Rgb2Gray');  
  
im4= imresize(im3, [167 250]);  
subplot(2, 2, 3),imshow(im4); title('Rgb-Resized');  
  
im5=edge(im4,'canny',[],0.5);  
subplot(2, 2, 4),imshow(im5); title('Gamma = 0.5');  
  
suptitle('Transforming Image to Find Edges');
```


Output Image - Transition

Transforming Image to Find Edges

lam2Double



Rgb2Gray



Rgb-Resized



Gamma = 0.5



Extracting Edges

```
im1=medfilt2(im1,[3 3]);  
BW = edge(im1,'sobel');  
[imx,imy]=size(BW);  
msk=[0 0 0 0 0; 0 1 1 1 0; 0 1 1 1 0; 0 1 1 1 0; 0 0 0 0 0];  
B=conv2(double(BW),double(msk));  
L = bwlabel(B,8);  
mx=max(max(L))  
[r,c] = find(L==17);  
rc = [r c];  
[sx sy]=size(rc);  
n1=zeros(imx,imy);  
for i=1:sx  
x1=rc(i,1);  
y1=rc(i,2);  
n1(x1,y1)=255;  
end  
figure,imshow(im);
```



Shape Extraction

```
figure,  
I_eq = adapthisteq(I_eq);  
subplot(2, 2, 1), imshow(I_eq)  
bw = im2bw(I_eq, graythresh(I_eq));  
subplot(2, 2, 2), imshow(bw)  
  
bw2 = imfill(bw, 'holes');  
bw3 = imopen(bw2, ones(5,5));  
bw4 = bwareaopen(bw3, 40);  
bw4_perim = bwperim(bw4);  
  
subplot(2, 2, 3), imshow(bw4_perim)  
mask_em = imextendedmax(I_eq, 30);  
subplot(2, 2, 4), imshow(mask_em)
```



OutPut Image - Comparison

```
figure,  
mask_em = imclose(mask_em, ones(5,5));subplot(2, 2, 1), imshow(mask_em)  
mask_em = imfill(mask_em, 'holes');subplot(2, 2, 2),imshow(mask_em)  
mask_em = bwareaopen(mask_em, 40);subplot(2, 2, 3),imshow(mask_em)
```

```
figure,  
I_eq_c = imcomplement(I_eq);  
subplot(2, 2, 4),imshow(I_eq_c)  
I_mod = imimposemin(I_eq_c, ~bw4 | mask_em);  
subplot(1, 2, 1),imshow(I_mod )  
L = watershed(I_mod);  
subplot(1, 2, 2),  
imshow(label2rgb(L))
```



Texture Extraction

```
I=imread('Auto2.jpg');  
figure,imshow(I)  
hy = fspecial('sobel');  
hx = hy';  
Iy = imfilter(double(I), hy, 'replicate');  
Ix = imfilter(double(I), hx, 'replicate');  
gradmag = sqrt(Ix.^2 + Iy.^2);  
figure, imshow(gradmag,[]),  
L = watershed(gradmag);  
figure, imshow(L)  
se = strel('disk', 20);  
Io = imopen(I, se);  
figure, imshow(Io), title('Opening (Io)')  
Ie = imerode(I, se);  
Iobr = imreconstruct(Ie, I);  
figure, imshow(Iobr)
```

Modified regional maxima superimposed on original image



Opening-closing by reconstruction (Iobrcbr)



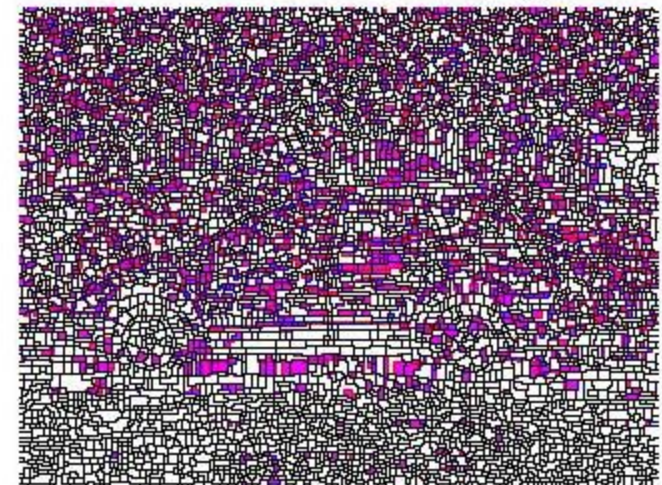
Texture Extraction

```
Ioc = imclose(Io, se);  
figure, imshow(Ioc)  
Iobrd = imdilate(Iobr, se);  
Iobrcbr = imreconstruct(imcomplement(Iobrd), imcomplement(Iobr));  
Iobrcbr = imcomplement(Iobrcbr);  
figure, imshow(Iobrcbr)  
fgm = imregionalmax(Iobrcbr);  
I2 = I;  
I2(fgm) = 255;  
figure, imshow(I2)  
se2 = strel(ones(5,5));  
fgm2 = imclose(fgm, se2);  
fgm3 = imerode(fgm2, se2);  
fgm4 = bwareaopen(fgm3, 20);  
I3 = I;  
I3(fgm4) = 255;  
figure, imshow(I3)
```

Lrgb superimposed transparently on original image



Colored watershed label matrix (Lrgb)



Texture Extraction

Thresholded



```
bw = im2bw(Iobrcbr, graythresh(Iobrcbr));  
figure, imshow(bw)  
D = bwdist(bw);  
DL = watershed(D);  
bgm = DL == 0;  
figure, imshow(bgm)  
L = watershed(gradmag);  
I4 = I;  
figure, imshow(I4)  
figure, imshow(L)  
figure, imshow(I), hold on  
himage = imshow(L);  
set(himage, 'AlphaData', 0.3);
```


SEPARATE IMAGE

```
a=imread('Auto2.jpg');  
b1 = im2double(a);  
b = rgb2gray(a);  
b1=10*round(b/10);  
figure,imhist(b);  
figure,imhist(b1);
```

```
figure,  
a2=255*((a>=0)&(a<=14));  
subplot(2, 2, 1),imshow(a2);  
a3=255*((a>=15)&(a<=15));  
subplot(2, 2, 2), imshow(a3);  
figure,  
a4=255*((a>=16)&(a<=99));  
subplot(2, 2, 1), imshow(a4);  
a5=255*((a>=100)&(a<=149));  
subplot(2, 2, 2), imshow(a5);  
a6=255*((a>=150)&(a<=220));subplot(2, 2, 3), imshow(a6);  
a7=255*((a>=221)&(a<=255));subplot(2, 2, 4), imshow(a7);
```



SEPARATE IMAGE

```
figure,  
b2=255*((b>=0)&(b<=14));  
subplot(2, 2, 1), imshow(b2);  
b3=255*((b>=15)&(b<=15));  
subplot(2, 2, 2), imshow(b3);
```



```
figure,  
b4=255*((b>=16)&(b<=99));  
subplot(2, 2, 1), imshow(b4);  
b5=255*((b>=100)&(b<=149));  
subplot(2, 2, 2), imshow(b5);  
b6=255*((b>=150)&(b<=220));  
subplot(2, 2, 3), imshow(b6);  
b7=255*((b>=221)&(b<=255));  
subplot(2, 2, 4), imshow(b7);
```



BACKGROUND EXTRACTED

